MSc Finance 20127

Corporate Finance (Business Valuation)
Special Topic 01 — Coherent Plans

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Where We Are in the Valuation Workflow

Equity story $\longrightarrow$ Financial statements $\longrightarrow$ **Forecasts** $\longrightarrow$ **Cash flows** $\longrightarrow$ Discount rate $\longrightarrow$
Terminal value $\longrightarrow$ Multiples $\longrightarrow$ Value bridge $\longrightarrow$ Recommendation

Today:
Forecasts $\longrightarrow$ **Cash Flows** $\longrightarrow$ **Value**

Why May Good-Looking Forecasts Still Produce Bad Valuations?

- Each assumption may look reasonable in isolation
- The problem is whether the assumptions fit together
- Valuation errors often come from inconsistency across revenue, margins, CapEx, working capital, debt, and terminal assumptions
- **Bottom line**: The unit of analysis is not a single forecast. It is the forecast system

Coherence Is Not The Same As Accuracy

	Accurate Ex Post	Inaccurate Ex Post
Coherent Ex Ante	Good process, Good outcome	Good process / Bad Luck / Bad outcome
Incoherent Ex Ante	Bad process, Lucky outcome	Bad process / Bad outcome / Predictable error

Main message: Accuracy is judged after realizations; Coherence is assessed before realizations

Rules of Thumb in Business Planning

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How CFOs and Analysts Often Forecast

1. Plain growth
   - Compute for each item the average of the 2-3 most recent annual growth rates
2. Percentage of sales
   - Assigning to each item the same growth rate as that forecasted for sales
3. Economies of scale
   - Estimate OLS regression of each item's growth on contemporaneous sales growth
   - The estimated regression intercept is the fixed component
   - The estimated slope times the forecasted sales growth is the variable component
4. Industry-based
   - Same as 3. above, but using only data on firms in the same industry
5. Disaggregated / multivariate
   - Estimate a 'kitchen sink' specification with all available variables

Least Coherent

Most Coherent

The Misleading Simplicity of "Everything Grows with Sales"

Percentage-of-sales Model		Coherence Questions
Sales growth forecast/target	+10%	Does demand, market share, or capacity support this?
COGS / SG&A growth forecast	+10%	Are there fixed costs or scale economies?
Net Working Capital growth forecast	+10%	Does working capital intensity remain constant?
Capital Expenditures growth forecast	+10%	Is there maintenance CapEx?

- Percentage-of-sales is not wrong by definition; It is wrong when it ignores the economics of the business

The Paper's Findings

Incoherence Is Not Just An Academic Concern

1. CFO forecasts often align with rules of thumb
2. Some rules are more incoherent than others
   - Plain growth and Percentage-of-sales most incoherent; Multivariate most coherent
   - Nearly half of CFO forecasts align with Plain growth and Percentage-of-sales RoTs
3. Incoherence is associated with:
   - predictable forecast errors
   - lower performance
   - underinvestment

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How Incoherent Plans Distort Valuation

Incoherent plan $\longrightarrow$ Wrong reinvestment $\longrightarrow$ wrong FCFO $\longrightarrow$ wrong terminal value $\longrightarrow$ Wrong Enterprise Value and Equity Value

Examples:

- Sales growth without CapEx $\rightarrow$ FCFO overstated
- Working capital flat despite growth $\rightarrow$ cash needs understated
- Debt/cash used as plug $\rightarrow$ financing policy hides operating inconsistency
- Terminal growth not reachable from explicit period $\rightarrow$ terminal value becomes fantasy
- Margin expansion without mechanism $\rightarrow$ terminal margin overstated

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Examples

JJ Jones: Which Assumptions Must Fit Together?

Income Statement

- Revenue growth — simple 10% growth
- Cost of goods sold — 60% of revenues
- Operating expenses — 10% of revenues

Assets

- Excess assets (cash and other) — $0
- Required cash balance — 5% of revenues
- Accounts receivable — 50 days*
- Inventory — 80 days*
- Net PPE to revenues — 1.2 ratio*

Liabilities and Preferred Stock

- Accounts payable — 40 days*
- Accrued expenses — 20% of revenues
- Debt — retire current portion of long-term debt, increase long-term if the company has insufficient cash flow (same interest rate)

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- What else must be true for these assumptions to describe one coherent business?
- JJ Jones is not just a modeling exercise; It is a coherence exercise

The Coherence Checklist

1. **Story-to-driver coherence**
   Does the equity story explain the forecast drivers?

2. **Revenue-to-capacity coherence**
   Does growth require CapEx, capacity, hiring, inventory, or marketing?

3. **Margin coherence**
   Do margin changes follow from price, mix, scale, efficiency, or input costs?

4. **Working-capital coherence**
   Do receivables, inventory, and payables reflect the business model?

5. **Financing coherence**
   Are debt, cash, dividends, and repurchases consistent with FCFE?

6. **Terminal coherence**
   Is the terminal firm reachable from the explicit forecast?

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Plausible is Not Necessarily Coherent

- Excel can make a model balance even when assumptions are economically inconsistent
- AI can generate plausible narratives and reasonable-looking assumptions

$\Longrightarrow$ *Neither Excel nor AI guarantees coherence*

- The analyst's job is to test the system
- **Bottom line**: A model that balances is not necessarily a business plan that makes sense

Discussion Question

Consider the following system of forecasts:

- Sales growth 12% per year
- EBITDA margin rises from 15% to 22%
- CapEx falls below depreciation
- NWC remains constant
- Debt is repaid quickly
- Terminal growth is 4%

i. Under what business story could this plan be coherent?

ii. Which assumptions are red flags absent that story?

Takeaways

1. Coherence is a property of the forecast system (plan), not one forecast variable
2. A plan can be plausible line by line and incoherent overall
3. Rules of thumb are useful only when they respect business economics
4. Incoherent plans distort FCFO, terminal value, leverage, and valuation
5. Good analysts test whether story, assumptions, financial statements, and valuation all fit together

In valuation, discipline is not only getting the formula right. It is making the assumptions fit together